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Thermodynamic descriptions of the Ag-X (X = S, As, Lu) systems

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ABSTRACT

Based on the critical evaluation of experimental data available in the literature, for the first time the Ag-X (X = S, As, Lu) binary systems were assessed thermodynamically using the CALPHAD (CALculation of PHAse Diagram) method. The solution phases including liquid, (Ag), (αS), (βS), (αAs), (εAs) and (Lu) were described by the substitutional solution model, and their excess Gibbs energies were described with the Redlich-Kister polynomial equation. In view of the possible existence of associate species Ag_2S and the difficulty to describe the liquid phase by the substitutional solution model, an associate model was used to describe the liquid phase in the Ag-S system. The intermetallic compounds, i.e. αAg_2S, AgLu, Ag_2Lu and Ag_4Lu, were all modeled as stoichiometric compounds. And the intermetallic compounds βAg_2S and γAg_2S were described by two-sublattice model $(Ag, S)_2S_1$ in view of the solubilities. Self-consistent thermodynamic parameters to describe the Gibbs energies of phases in the Ag-S, Ag-As and Ag-Lu systems were obtained. The calculated results were in good agreement with the reported phase equilibria and thermodynamic properties.

Introduction

Materials with a high addition of Ag are widely used in electrical contact and welding field as they display superior electrical property and thermal conductivity. In practice, two main groups can be divided of these materials as Ag compounds and Ag-alloys based on its phase constitution, respectively [1]. For example, the compound Ag_2S is a very useful compound and has been used in many applications, including photovoltaic cells, photoconductors, photosensitive films in optoelectronic devices, solar-selective coatings and infrared detectors [2,3]. Moreover, the hollow Ag_2S sphere with a strong absorption in ultraviolet and infrared region and weak in visible region has been confirmed with unique optical properties [4]. Limited investigation about the Ag-As system was reported in the literature. It was found [5,6] that As can not only cause severe environmental pollution, but also produce serious adverse effect on human health. However, As aptamer can be served by absorbing on sliver as a SERS (Surface-enhanced Raman scattering) donor [7]. In view of this, the investigation of Ag-As system is valuable to medicine field. It is well known that most Ag-based alloys are of poor strength and cannot withstand high temperature. Fortunately, addition of some RE (rare earth) elements into Ag-based alloys may help to achieve high strength and improve their performances at elevated temperatures [8–10]. Besides, the addition of some RE elements, like Lu element, will improve thermopowers accompanied with residual resistivities in comparison with non-transition metal impurities with the same valence difference [11]. Many Ag-RE systems [12,13] have been assessed thermodynamically. But this is not the case for the Ag-Lu system. The design of compositions of Ag-based novel materials requires information about the phase equilibria and thermodynamic properties in the Ag-based systems. The CALPHAD method is a useful tool to develop materials with less cost and time. It is well known that the establishment of a thermodynamic database for multi-component alloys requires reasonable thermodynamic descriptions of the binary subsystems. The purposes of the present work are (1) to critically review all the available experimental data for the Ag-X (X = S, As, Lu) systems, and (2) to obtain a set of self-consistent thermodynamic parameters for each of these binary systems by means of the CALPHAD method.

Evaluation of experimental data

All the crystal structure data in the Ag-X (X = S, As, Lu) systems are

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Table 1
Crystal structure data for the solid phases in the Ag-X (X = S, As, Lu) systems.

System	Phase	Person symbol	Space group	Prototype	Lattice parameter (Å)			Reference
					a	*b*	*c*	
Ag-S	(Ag)	*cF4*	$Fm\bar{3}m$	Cu	0.4086	–	–	[14]
	αAg_2S (tetragonal)	*mP24*	$P2_1/c$	–	0.953	0.692	0.828	[15]
	αAg_2S (acanthite)	*mP12*	$P2_1/n$	–	0.423	0.691	0.787	[16]
	βAg_2S	*cI20*	$Im\bar{3}m$	–	0.4864	–	–	[17]
	γAg_2S	*cF44*	$Fm\bar{3}m$	Cu_2Se	0.634	–	–	[18]
	(αS)	*oF128*	*Fddd*	αS	1.0465	1.2866	2.4486	[14]
	(βS)	*mP48*	$P2_1/a$	–	1.092	1.098	1.104	[19]
Ag-As	ζ	*hP2*	$P6_3/mmc$	Mg	0.2899	–	0.4734	[20]
	(αAs)	*hR2*	$R\bar{3}m$	αAs	0.4132	–	–	[14]
	(εAs)	*oC8*	*Cmca*	αGa	0.362	1.085	0.448	[21]
Ag-Lu	(Lu)	*hP2*	$P6_3/mmc$	Mg	0.35052	0.55494	9.841	[22]
	AgLu	*cP2*	*Pm3m*	CsCl	0.3541	–	–	[23]
	Ag_2Lu	*tI6*	*I4/mmm*	$MoSi_2$	0.3636	–	0.9084	[24]
	Ag_4Lu	*tI10*	*I4/m*	$MoNi_4$	0.6670	–	0.4158	[25]

Table 2
Summary of the experimental data on phase equilibria in the Ag-X (X = S, As, Lu) systems.

System	Rel.	Experimental technique	Temperature/composition/phase range studied	QM[a]
Ag-S	[26]	Thermal analysis and differential thermal analysis	700–950 °C, 10–100 at% S	+
	[27]	Thermal analysis, the decomposition measurements with the H_2S/H_2 method	Homogeneity range of the Ag_2S	–
	[28]	Thermal analysis, coulometric method, gas equilibration method and X-ray diffraction	600–1200 °C, 5–33 at% S, Solubility limits of S in (Ag), liquid phase boundary	+
	[29]	Thermal analysis, EMF method using solid oxide electrolyte	594–1265 °C, 0–33 at% S, phase boundary	–
	[30]	Thermal analysis, microscopic methods and X-ray diffraction	900–1000 °C, 0–15 at% S, phase boundary	+
Ag-As	[21]	Thermal analysis, microscopic methods and X-ray diffraction	300–1000 °C, 0–20 at% As, phase boundary	+
	[31]	Thermal analysis and microscopic methods	300–1000 °C, 0–100 at% As, Solubility of As in (Ag)	+
Ag-Lu	[32]	Thermal analysis and X-ray parametric method	300–814 °C, 0–20 at% Lu, Solubility of Lu in (Ag)	+
	[33]	DTA in alumina crucible under argon	The melting temperature of Ag_2Lu	+

[a] QM means Quoted Mode, which indicates whether the data are used in the parameter optimization: + used; - not used.

summarized in Table 1 according to the literature data [13–25]. In order to facilitate reading, the experimental data on the phase diagram [21,26–33] of these systems are briefly summarized with the quoted mode or each reference in Table 2. Each of the binary systems is critically reviewed as follows.

The Ag-S system

The Ag-S phase diagram over the entire composition range was first investigated by Kracek [26] using differential thermal analysis (DTA) and thermal analysis (TA). The transition temperatures of $\alpha Ag_2S \rightarrow \beta Ag_2S$, $\beta Ag_2S \rightarrow \gamma Ag_2S$ and $\gamma Ag_2S \rightarrow$ liquid were determined to be 178, 571 and 838 °C, respectively [26]. The crystal structure data about αAg_2S, βAg_2S and γAg_2S were investigated by Djurle and An [15], Shafizade et al. [16], Dzhabrailova and Asadov [17], and Frueh [18], as shown in Table 1. Two monotectic reactions, $L_1 \rightarrow (Ag) + L_2$ and $L_2 \rightarrow \gamma Ag_2S + L_3$, were determined to be 906 °C at 5.8 at% S as well as 740 °C at 36 at% S, respectively. Two eutectic reactions, $L_2 \rightarrow (Ag) + \gamma Ag_2S$ and $L_3 \rightarrow \alpha Ag_2S + (\beta S)$ were determined to be at 804 °C and 32 at% S as well as 119 °C and 99.99 at% S, respectively.

Rau [27] performed sulphur fugacity measurements within the homogeneity range of Ag_2S in the temperature range between 465 and 747 °C. The compound Ag_2S is treated as a stoichiometric compound in this work in view of negligible homogeneity range. The solubility limits of S in (Ag) at 604, 656, 706 and 752 °C were approximately determined by the coulometric method [28]. In the solid state, S exists in two allotropic forms-(αS), which is stable up to 95.5 °C and (βS) being stable from 95.5 °C to its melting point at 115.22 °C [34]. The reliable solubilities of Ag in (αS) and (βS) have not been determined. However, they are most likely to be negligible according to the work of Sharma and Chang [34]. The activity of S in liquid Ag-S alloys up to 33 at% S has been determined by Rosenqvist [28] using the gas equilibration method in the temperature range from 800° to 1250°C.

The enthalpy of formation at room temperature for Ag_2S has been determined by Keyes and Felsing [35], Kiukkola and Wagner [36], Bartholomew et al. [37], Piacente and Scardala [38] as well as Bryndzia and Kleppa [39]. It should be mentioned that Bryndzia and Kleppa [39] determined the enthalpy of formation for Ag_2S at room temperature to be -22.4 ± 0.5 kJ/(mol of formula), which is too smaller than the other researchers' results. Consequently, the data from Bryndzia and Kleppa [39] are excluded from the present optimization. The enthalpies of formation of the Ag_2S phase were calculated by means of the first-principles calculations, which are included in the Open Quantum Materials Database [40] and the Materials Project [41]. The above measured and calculated enthalpies of formation for Ag_2S are summarized in Table 3. The enthalpies of transformation for $\alpha Ag_2S \rightarrow \beta Ag_2S$ ($\Delta H_{\alpha-\beta}$), $\beta Ag_2S \rightarrow \gamma Ag_2S$ ($\Delta H_{\beta-\gamma}$) were measured by Thompson and Flengas [42] to be 1.33 and 0.17 kJ/mole-atoms, respectively. Ditman and Kulikova [43] reported the enthalpy of melting of γAg_2S to be -3.62 kJ/mole-atoms by the dew point method. Recently, Sadovnikov et al. [44] determined the transition enthalpy for the αAg_2S (acanthite) $\rightarrow \beta Ag_2S$ phase transformation to be 1.2–1.3 kJ/mole-atoms at 176 °C by energy dispersive X-ray analyzer and DTA-DTG (derivative thermogravimetric) method. The results of Sadovnikov et al. [44] are generally consistent with that reported by Thompson and Flengas [42]. Thus, the data of Thompson and Flengas [42] were utilized in the present optimization. Besides, Fukatsu and Kozuka [29] as well as Friedrich and Lerouz [30] have experimentally determined the liquidus of Ag-S between 0 and 33 at% S. Their results were also used in the present thermodynamic optimization.

Table 3
Summary of experimental and previously assessed data used for the enthalpies of formation at room temperature.

Compound	$\Delta_f H(298K)$ kJ/mole-atoms	Method	Reference
Ag_2S	− 10.1	Solution calorimeter	[35]
	− 9.5 ± 0.2	Electromotive force of solid-state galvanic cells	[36]
	− 10.5	Direct synthesis calorimetry	[37]
	− 10.7	Vapor pressure calculations	[38]
	− 7.5 ± 0.2	Drop calorimetry	[39]
	− 14.7 (*P21/m*)	First-principles calculations	[40]
	− 15.2 (*P21*)	First-principles calculations	[40]
	− 26.8 (*P21*)	First-principles calculations	[41]
	− 26.3 (*P21/m*)	First-principles calculations	[41]
	− 24.6 (*P21/c*)	First-principles calculations	[41]
	− 18.8 (*P21*)	First-principles calculations	[41]
	− 10.9	Assessed	This work
AgLu	− 41.3 ± 1.8	High-temperature direct synthesis calorimetry	[52]
	− 34.3(*Pm3m*)	First-principles calculations	[40]
	− 41.0	Assessed	This work
Ag_2Lu	− 25.6 ± 1.6	High-temperature direct synthesis calorimetry	[53]
	− 33.6 (*I4/mmm*)	First-principles calculations	[40]
	− 32.9 (*I4/mmm*)	First-principles calculations	[41]
	− 32.4	Assessed	This work
Ag_4Lu	− 23.9 ± 4.2	High-temperature direct synthesis calorimetry	[52]
	− 22.5 (*I4/m*)	First-principles calculations	[40]
	− 22.3 (*I4/m*)	First-principles calculations	[41]
	− 23.9	Assessed	This work

The Ag-As system

The Ag-As phase diagram was first determined by Heike and Leroux [31] using thermal analysis and microscopic techniques. They also determined the solid solubility of As in (Ag) to be about 7 at% from 300° to 595 °C. However, Owen et al. [45,46] later performed some experimental work by XRD technique and found that the solubility of As in Ag increased from about 4.5 at% at 300 °C to about 8.8 at% at 595 °C. This is different from the results of Heike et al. [31]. Eade and Rothery [20] determined the phase diagram with the composition range from 0 to 20 at% As using thermal analysis, microscopic and XRD techniques. Additionally, the intermediate phase ζ ranging from 9.6 to 11 at% As was determined by Eade and Rothery [17]. They also reported the lattice parameter-composition curves of the single-phase alloys by means of XRD and chemical analysis. The same method was used to determine the [(Ag) + ζ]/ζ and ζ/[ζ + (As)] phase boundaries. Based on XRD measurements, Broderick and Ehret [47] concluded that Ag was practically insoluble in solid As. The eutectic reaction L → ζ + (As), the eutectoid reaction ζ→ (Ag) + (As) and the peritectic reaction L + (Ag) → ζ were determined to be at 540 °C and 25.3 at% As, 446 °C and 9.5 at% As as well as 582 °C and 10 at% As, respectively. Broderick and Ehret [47] also found the existence of ζ with hcp_A3 structure. However, no enthalpy of formation data has been reported for ζ phase.

The experimental data about the thermodynamic activity of As in (Ag) solid solutions from 518° to 659°C were determined by Ludecke et al. [48] using vapor pressure measurements. These data were utilized but with a low weight during the present optimization because the experimental data were too scattered.

The Ag-Lu system

Gschneidner et al. [32] determined the solid solubility of Lu in (Ag) to be 5.78 at% from 300° to 814°C by thermal analysis and X-ray parametric method. The eutectic reaction L → Ag_4Lu + (Ag) was determined to be at 814 °C. And this eutectic composition may be at 12 at % Lu according to Gschneidner et al. [32]. The crystal structure data about AgLu, Ag_2Lu and Ag_4Lu were investigated by Dwight [23], Iandelli and Palenzona [24], and Mcmasters et al. [25], as shown in Table 1. The melting temperature of Ag_2Lu was determined to be at 1100 °C by Stapf and Jehn [33] using DTA in alumina crucible under the atmosphere of argon. The characteristics of the Ag-Lu system were in analogy to other Ag-RE (RE = Er, Ho and Y) [49–51] systems. Due to the limited experimental data of the Ag-Lu system available in the literature, the invariant reactions, i.e. L → Ag_4Lu + Ag_2Lu, L → Ag_2Lu + AgLu and L → AgLu + (Lu), have been assumed based on the compounds in the Ag-Lu system and the similarity between the Ag-Lu system and other Ag-RE (RE = Er, Ho and Y) systems.

Fitzner and Kleppa [52] and Meschel and Kleppa [53] reported the standard enthalpies of formation of the intermetallic compounds AgLu and Ag_4Lu, and Ag_2Lu by direct synthesis in a high-temperature calorimeter, respectively. Furthermore, the enthalpies of formation of the AgLu, Ag_2Lu and Ag_4Lu phases were also calculated by means of the first-principles calculations [40,41]. These experimental data and first-principles calculated results are also listed in Table 3. The experimental results [52,53] are used in the present thermodynamic optimization.

Thermodynamic models

Pure elements

The Gibbs energy function ${}^0G_i^{\varphi}(T) = G_i^{\varphi} - H^{SER}$ for the pure elements i (i = Ag, S, As, Lu) in the phase φ can be expressed by the following equation [51]:

$$ {}^0G_i^{\varphi} = a + bT + cT\ln(T) + dT^2 + eT^2 + fT^{-1} + gT^7 + hT^{-9} \quad (1) $$

in which H_i^{SER} represents the mole enthalpy of element at room temperature and 1 bar in its standard element reference (SER) state, and T is the absolute temperature. The last two terms in Eq. (1) are used only outside the ranges of stability [52], the term gT^7 is for a liquid below the melting and hT^{-9} is for solid phase above the melting point. In the present modeling, the Gibbs energies as a function of temperature are taken from the SGTE (Scientific Group Thermodata Europe) compilation by Dinsdale [54].

Solution phases

In the Ag-X (X = S, As, Lu) binary systems (A-B), the solution phases, i.e., liquid in the Ag-As and Ag-Lu system, (Ag), (αS), (βS), (αAs), (εAs) and (Lu), are described by the substitutional solution model. The Gibbs energy can be described by the Redlich-Kister polynomial [55]:

$$ G_m^{\phi} = \sum_{i=\mathrm{A,B}} {}^0G_i^{\phi} x_i + RT \sum_{i=A,B} x_i \ \mathrm{In}\ x_i + x_A x_B \sum_{j=0}^{n} {}^jL_{A,B}^{\phi}(x_A - x_B)^j \quad (2) $$

where ${}^0G_i^{\phi}$ represents the Gibbs free energy of the pure component i (i = Ag, S, As, Lu) in the reference state with ϕ phase, x_i is the mole fraction of component, R is the gas constant, T *is* the absolute temperature, and $L_{\mathrm{A,B}}^{\phi}$ is the binary interaction parameter, which can be expressed as:

$$ {}^jL_{\mathrm{A,B}}^{\phi} = a_j + b_j T \quad (3) $$

The interaction coefficients a_j and b_j (j = 0, 1, 2…) are to be evaluated in the optimization process. Similar expressions like Eqs. (2) and (3) can be applied to describe the ζ phase with hcp_A3 structure in the Ag-As system.

Associate model

When very strong attractive interactions exist in some liquids, it is likely that there will be a tendency for short-range ordering around

Table 4
Summary of the thermodynamic parameters in the Ag-X (X = S, As, Lu) systems[a].

System	Phase	Sublattice model	Evaluated parameters
Ag-S	Liquid	$(Ag, Ag_2S, S)_1$	${}^0G^{liq.}_{Ag_2S} = -24814 - 30.38T + 2G^{liq.}_{Ag} + G^{liq.}_{S}$
			${}^0L^{liq.}_{Ag,Ag_2S} = 14740 + 5.86T$
			${}^1L^{liq.}_{Ag,Ag_2S} = 31246 - 19.97T$
			${}^0L^{liq.}_{Ag_2S,S} = 69068 + 3.33T$
			${}^1L^{liq.}_{Ag_2S,S} = -62051 - 17.21T$
			${}^2L^{liq.}_{Ag_2S,S} = 26453 - 57.33T$
	αAg_2S	$(Ag)_{0.6667}(S)_{0.3333}$	${}^0G^{\alpha Ag_2S}_{Ag:S} = -10900 - 9.44T$
			$+ 0.6667G^{Fcc_A1}_{Ag} + 0.3333G^{Orth}_{S}$
	βAg_2S	$(Ag, S)_{0.6667}(S)_{0.3333}$	${}^0G^{\beta Ag_2S}_{Ag:S} = -10598 - 10.11T$
			$+ 0.6667G^{Fcc_A1}_{Ag} + 0.3333G^{Orth}_{S}$
			${}^0G^{\beta Ag_2S}_{S:S} = 12157 - 1.79T + G^{Orth}_{S}$
			${}^0L^{\beta Ag_2S}_{Ag,S:S} = -6100 + 20.7T$
	γAg_2S	$(Ag, S)_{0.6667}(S)_{0.3333}$	${}^0G^{\gamma Ag_2S}_{Ag:S} = -10300 - 10.46T$
			$+ 0.6667G^{Fcc_A1}_{Ag} + 0.3333G^{Orth}_{S}$
			${}^0G^{\gamma Ag_2S}_{S:S} = 8743 + 24.07T + G^{Orth}_{S}$
			${}^0L^{\gamma Ag_2S}_{Ag,S:S} = -6459 + 71.37T$
Ag-As	Liquid	$(Ag, As)_1$	${}^0L^{liq.}_{Ag,As} = 97014 - 116.19T$
			${}^1L^{liq.}_{Ag,As} = -99005 + 88.22T$
	Fcc_Al	$(Ag, As)_1$	${}^0L^{Fcc_A1}_{Ag,As} = 70722 - 67.84T$
			${}^1L^{Fcc_A1}_{Ag,As} = -61395 + 41.64T$
	ζ	$(Ag,As)_1$	${}^0L^{\zeta}_{Ag,As} = 31429 - 0.10T$
			${}^1L^{\zeta}_{Ag,As} = -18337 - 45.10T$
Ag-Lu	Liquid	$(Ag, Lu)_1$	${}^0L^{liq.}_{Ag,Lu} = 157518 + 86.90T$
			${}^1L^{liq.}_{Ag,Lu} = -4888 - 7.25T$
	Fcc_A1	$(Ag, Lu)_1$	${}^0G^{Fcc_A1}_{Ag,Lu} = 75657 + G^{Hcp_A3}_{Lu}$
			${}^0L^{Fcc_A1}_{Ag,Lu} = -138640$
			${}^1L^{Fcc_A1}_{Ag,Lu} = 6398$
	Hcp_A3	$(Ag, Lu)_1$	${}^0L^{Hcp_A3}_{Ag,Lu} = 30000$
	AgLu	$(Ag)_{0.5}(Lu)_{0.5}$	${}^0G^{AgLu}_{Ag:Lu} = -41000 + 18.34T$
			$+ 0.5G^{Fcc_A1}_{Ag} + 0.5G^{Hcp_A3}_{Lu}$
	Ag_2Lu	$(Ag)_{0.6667}(Lu)_{0.3333}$	${}^0G^{Ag_2Lu}_{Ag:Lu} = -32410 + 11.95T$
			$+ 0.6667G^{Fcc_A1}_{Ag} + 0.3333G^{Hcp_A3}_{Lu}$
	Ag_4Lu	$(Ag)_{0.8}(Lu)_{0.2}$	${}^0G^{Ag_4Lu}_{Ag:Lu} = -23900 + 8.88T$
			$+ 0.8G^{Fcc_A1}_{Ag} + 0.2G^{Hcp_A3}_{Lu}$

[a] Gibbs energy in J/mol-atom, temperature (T) in Kelvin and pressure (P) in *Pa*. The Gibbs energies for the pure elements are from the SGTE compilation [54].

certain compositions. One can therefore hypothesize the existence of associates, which are molecules with definite stoichiometries but which exhibit short life time and cannot be isolated as independent entities [56]. In the Ag-S system, two miscibility gaps in the liquid phase exits on Ag-rich and S-rich sides of the compound Ag_2S and indicate that a very strong attractive interactions exist in the liquid phase. The short-range ordering may occur in the liquid phase. There is no this phenomenon of the liquid phase in the Ag-As and Ag-Lu systems. Thus, the associate model [57] was adopted to describe Ag-S liquid and the substitutional solution model was used to describe Ag-As and Ag-Lu liquid. In addition, the associated specie is selected as Ag_2S because it is the compound with the highest melting temperature and the only one that melts congruently.

In the Ag-S system, the liquid phase is assumed to be composed of three species in the associate model (Ag, S, $Ag_2S)_1$. The molar Gibbs energy is expressed as follows:

$$G^{liq.}_m = y_{Ag}G^{liq.}_{Ag}(T) + y_S G^{liq.}_S(T) + y_{Ag_2S}G^{liq.}_{Ag_2S}(T) + RT\left(y_{Ag}lny_{Ag} + y_S lny_S + y_{Ag_2S}lny_{Ag_2S}\right) + {}^EG^{liq.}_m \quad (4)$$

where $G^{liq.}_{Ag_2S}(T)$ is the molar Gibbs energy of Ag_2S in the liquid phase, and ${}^EG^{liq.}_m$ can be calculated by the following equations:

$$G^{liq.}_{Ag_2S}(T) = 2G(\text{liquid, Ag}) + G(\text{liquid, S}) + a + bT \quad (5)$$

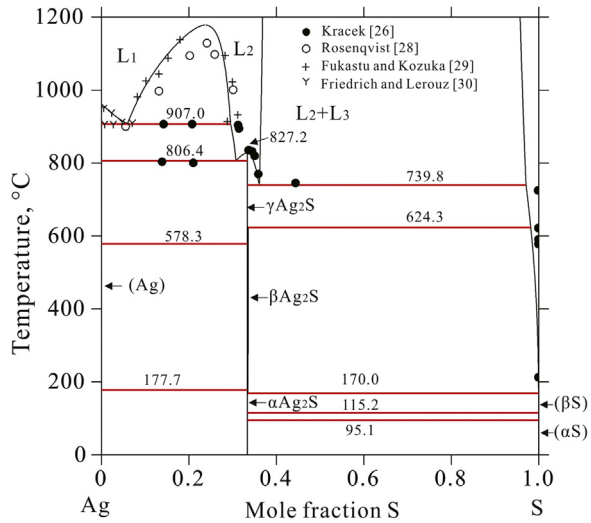

Fig. 1. Calculated Ag-S phase diagram along with the experimental data [26,28–30].

$$^{E}G_m^{\text{liq.}} = y_{\text{Ag}}y_{\text{S}}\sum_{i=0}^{n} {}^{i}L_{\text{Ag,S}}^{liq.}(y_{\text{Ag}} - y_{\text{S}})^i + y_{\text{Ag}}y_{\text{Ag}_2\text{S}}\sum_{i=0}^{n} {}^{i}L_{\text{Ag,Ag}_2\text{S}}^{liq.}(y_{\text{Ag}} - y_{\text{Ag}_2\text{S}})^i + y_{\text{Ag}_2\text{S}}y_{\text{S}}\sum_{i=0}^{n} {}^{i}L_{\text{Ag}_2\text{S,S}}^{liq.}(y_{\text{Ag}_2\text{S}} - y_{\text{S}})^i \quad (6)$$

where y_i represents the mole fraction of i species (i = Ag, S or Ag_2S), respectively.

Intermetallic compounds

The six compounds (αAg_2S, AgLu, Ag_2Lu and Ag_4Lu) are treated as stoichiometric compounds. The Gibbs energy of each compound, taking αAg_2S as an example, can be written as

$$G_m^{\alpha\text{Ag}_2\text{S}} - 2H_{\text{Ag}}^{\text{SER}} - H_{\text{S}}^{\text{SER}} = a^{\alpha\text{Ag}_2\text{S}} + b^{\alpha\text{Ag}_2\text{S}}T + 2{}^{0}G_{\text{Ag}}^{\text{Fcc_A1}} + {}^{0}G_{\text{S}}^{\text{Orth}} \quad (7)$$

in which $a^{\alpha Ag_2S}$ and $b^{\alpha Ag_2S}$ are the parameters to be evaluated during the optimization process. ${}^{0}G_{Ag}^{Fcc_A1}$ and ${}^{0}G_{S}^{Orth}$ are the mole Gibbs energy of Ag and S in their reference states. Similar expressions like Eq. (7) can be applied to describe the AgLu, Ag_2Lu and Ag_4Lu phases.

The intermetallic compounds βAg_2S and γAg_2S with a homogeneity range were described by a two-sublattice model [58,59] $(Ag, S)_2(S)_1$. The Gibbs energies per mole of formula unit Ag_2S is expressed as follows:

$$G_{\text{m}}^{\text{Ag}_2\text{S}} - H^{SER} = y_{\text{Ag}}G_{\text{Ag:S}}^{\text{Ag}_2\text{S}} + y_{\text{S}}G_{\text{S:S}}^{\text{Ag}_2\text{S}} + 2RT(y_{\text{Ag}}\ln y_{\text{Ag}} + y_{\text{S}}\ln y_{\text{S}}) + y_{\text{Ag}}y_{\text{S}}\sum_{i} {}^{i}L_{\text{Ag,S:S}}^{\text{Ag}_2\text{S}}(y_{\text{Ag}} - y_{\text{S}})^i \quad (8)$$

in which

$${}^{0}G_{\text{Ag:S}}^{\text{Ag}_2\text{S}} = \text{A}_0 + \text{B}_0T + 2{}^{0}G_{\text{Ag}}^{\text{Fcc_A1}} + {}^{0}G_{\text{S}}^{\text{Orth}} \quad (9)$$

$${}^{0}G_{\text{S:S}}^{\text{Ag}_2\text{S}} = \text{A}_1 + \text{B}_1T + 3{}^{0}G_{\text{S}}^{\text{Orth}} \quad (10)$$

$${}^{i}L_{\text{Ag,S:S}}^{\text{Ag}_2\text{S}} = \text{A}_2 + \text{B}_2T \quad (11)$$

where y_{Ag} and y_S are the site fractions of Ag and S in the first sublattice, respectively; $G_{Ag:S}^{Ag_2S}$ and $G_{S:S}^{Ag_2S}$ represent the end-members Ag_2S and S. The coefficients A_j and B_j (j = 0, 1, 2…) are to be evaluated according to the experimental data.

Results and discussion

Based on the reported experimental phase diagram and thermodynamic data, the optimization of the thermodynamic parameters was carried out by the computer-operated optimization program PARROT module of the Thermo-Calc software package developed by Sundman et al. [60]. The step-by-step optimization procedure carefully described by Du et al. [61] was utilized in the present work. The experimental phase diagram data and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the uncertainties of the experimental data. These uncertainties may be changed by trial and error during the assessment until most of the selected experimental information was reproduced within the expected uncertainty limits.

Taking the Ag-S system as an example, the thermodynamic parameters of the Ag-S system obtained using the associate model is listed in Table 4. The phase diagram calculated by the associate model is in a better agreement with the experimental data than the one calculated by the substitutional solution model, as discussed in Section 3.3. Therefore, only the optimization procedure by using the associate model is shown in the present work.

The optimization began with αAg_2S phase. The experimental enthalpy of formation [38] of Ag_2S was used to evaluate the value of coefficient $a^{\alpha Ag_2S}$ in Eq. (7). And the parameters $a^{\beta Ag_2S}$ and $a^{\gamma Ag_2S}$ in Eq. (7) for βAg_2S and γAg_2S were adjusted to reproduce the phase diagram data [42]. Then, the parameters a and b in Eq. (5) for liquid phase were adjusted using experimental information associated with the Ag_2S phase. The activity data of S in liquid Ag-S alloys as a function of composition at different temperatures [28] were also reproduced. In the next step, the $b^{\alpha Ag_2S}$, $b^{\beta Ag_2S}$ and $b^{\gamma Ag_2S}$ parameters for αAg_2S, βAg_2S and γAg_2S compounds were evaluated to describe the four invariant reactions and the phase relations associated with them. It was found that introducing b for the liquid phase could describe the remaining liquidus data. During modeling the Ag-S system, it was found that ${}^{i}L_{Ag,Ag_2S}^{liq.}$ has a large influence on the Ag-rich liquid phase and ${}^{i}L_{Ag_2S,S}^{liq.}$ large impact on the S-rich liquid. Therefore, the liquid parameters in these two regions were adjusted separately. Finally, all the parameters were optimized simultaneously using the reported experimental data. The optimization for the other two systems (Ag-As and Ag-Lu systems) was analogy with that for the Ag-S system. However, in the Ag-As and Ag-Lu systems, a substitution solution model was used to describe the liquid phase rather than the associate model. The finally assessed thermodynamic parameters for these systems are listed in Table 4.

Fig. 1 shows the calculated Ag-S phase diagram along with the experimental data [26,28–30]. As presented in Fig. 1, the calculated results in this work are in good agreement with the experimental data [26,28–30]. Fig. 2 presents the measured activities of S in liquid Ag-S alloys as a function of composition at 1000, 1100, and 1200 °C compared with the calculated activity. As shown in Fig. 2, the calculated three curves are consistent well with experimental data from Rosenqvist

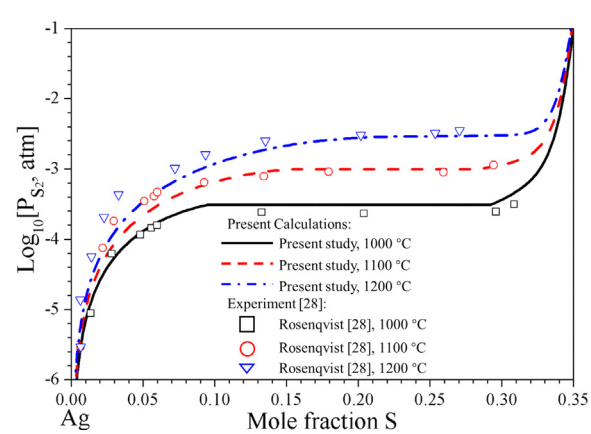

Fig. 2. Calculated S partial pressure in Ag-S liquid alloy compared with the experimental data [28].

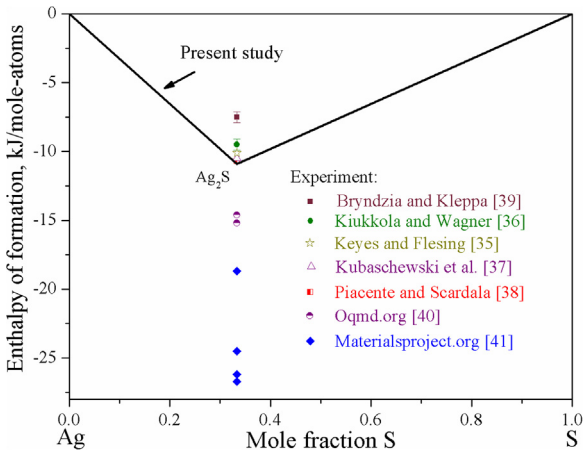

Fig. 3. Calculated enthalpies of formation at room temperature in the Ag–S system, compared with the experimental data [35–41].

[28]. The calculated enthalpies of formation in the Ag-S system compared with the experimental data and first-principles calculations are shown in Fig. 3. Since the enthalpies of formation for the Ag_2S phase obtained by different experiment methods [35–39] are very close to each other and the values from first-principles calculations [40,41] are much lower than the experimental values [35–39], the experimental results [35–39] are considered to be reasonable and thus accepted in the present optimization. As can be seen from Table 3 and Fig. 3, the calculated enthalpy of formation for the Ag_2S phase is in good agreement with the experimental results [35–39] in view of the experiment error. The calculated transition enthalpies of $\alpha Ag_2S \rightarrow \beta Ag_2S$ ($\Delta H_{\alpha-\beta}$) and $\beta Ag_2S \rightarrow \gamma Ag_2S$ ($\Delta H_{\beta-\gamma}$) in this work are 1.55 and 0.17 kJ/mole-atoms, respectively. As can be seen, the calculated values for $\Delta H_{\alpha-\beta}$ and $\Delta H_{\beta-\gamma}$ are in coincident with the experimental results (1.33 and 0.17 kJ/mole-atoms) determined by Thompson and Flengas [42].

Table 5 summarizes the computed temperatures and compositions of the invariant reactions in the Ag-S binary system along with the experimental data [21,26,30]. The agreement between the calculated results and experimental data is reasonable within the experimental errors. The calculated temperatures of reactions $\gamma Ag_2S + L_3 \rightarrow \beta Ag_2S$ and $\gamma Ag_2S \rightarrow (Ag) + \beta Ag_2S$ are 624.3 and 578.3 °C, which agree well with the experimental results (622 and 571 °C, respectively). And the calculated reaction temperatures for $L_3 + \beta Ag_2S \rightarrow \alpha Ag_2S$ and $\beta Ag_2S \rightarrow (Ag) + \alpha Ag_2S$ are 177.7 and 170.0 °C. These calculated results are consistent with experimental ones (176.0 and 178.0 °C, respectively).

The calculated Ag-As phase diagram in comparison with the experimental data [20,31] is shown in Fig. 4. It is evident that the calculated phase equilibria in this work agrees with the experimental data [20,31]. The computed phase diagram consists of three invariant reactions: $L \rightarrow \zeta + (As)$, $\zeta \rightarrow (Ag) + (As)$ and $L + (Ag) \rightarrow \zeta$ at 531, 488 and 589 °C, respectively, which are in good agreement with the experimental data [31]. Table 6 summarizes the computed invariant reactions in the Ag-As binary system along with the experimental data [31].

Fig. 5 presents the calculated activities of As in Ag-As alloys at 518.7, 565.2, 611 and 659 °C together with the experimental data [48]. Obviously, the calculated results with high As composition range at 659 °C agree with the experimental data, while the others seemingly cannot match well with the experimental data. It should be mentioned that the experimental activity data were obtained from the pseudoisopiestic method [48]. Through this method, the chemical excess potential is positive and shows weak minima during experimental measurement [48], indicating that a strong repulsive force. And this influence becomes more evident with the content of As increasing. In other words, there are some deviations or inaccuracy in the experimental results. If those activity data are reproduced well, the fcc phase region

Table 5
Calculated invariant reactions compared with the literature data in the Ag-S systems.

Reactions	Type	*T*, (°C)	Composition (at% *S*)			Reference
L1 ↔ (Ag) + L2	monotectic	906.0	5.8	–	31.1	[26] (exp.)
		905.0	5.8–6.2	–	30.8	[30] (exp.)
		907.0	5.9	0	29.7	This work (cal.)
L2 ↔ (Ag) + γAg2S	eutectic	804.0 ± 2	32	–	–	[26] (exp.)
		806.4	30.8	0	33.3	This work (cal.)
L2 ↔ γAg2S + L3	monotectic	740.0	36	–	99.9	[26] (exp.)
		739.8	36.2	33.3	97.2	This work (cal.)
L3 ↔ αAg2S + (βS)	eutectic	119.0	100	–	–	[26] (exp.)
		115.2	100	33.3	100	This work (cal.)
(βS) ↔ αAg2S + (αS)	eutectoid	95.0	100	33.3	100	[21] (cal.)
		95.1	100	33.3	100	This work (cal.)
γAg2S + L3 ↔ βAg2S	peritectic	622.0	33.4	99.9	33.5	[26] (cal.)
		624.3	33.3	98.5	33.8	This work (cal.)
γAg2S ↔ (Ag) + βAg2S	eutectoid	571.0	33.3	0.03	> 33.3	[26] (cal.)
		578.3	33.3	0	33.3	This work (cal.)
βAg2S + L3 ↔ αAg2S	peritectic	178.0	33.3	99.9	33.3	[26] (cal.)
		170.0	33.6	99.9	33.3	This work (cal.)
βAg2S ↔ (Ag) + αAg2S	eutectoid	176.0	33.3	0	33.3	[26] (cal.)
		177.7	33.3	0	33.3	This work (cal.)

exp. = experiment; cal. = calculation.

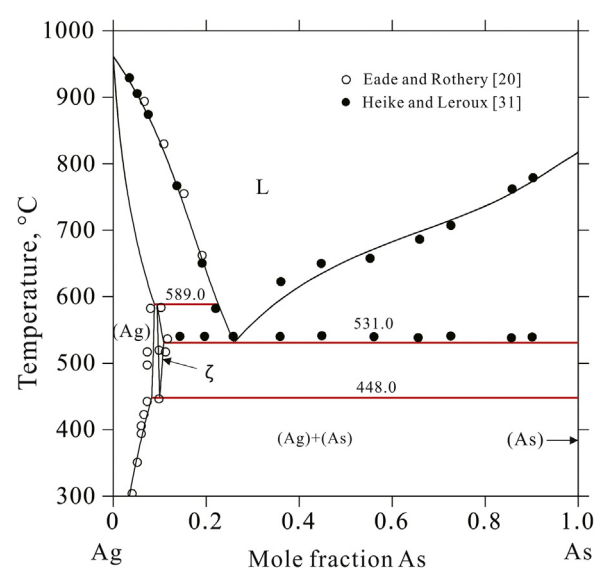

Fig. 4. Calculated Ag-As phase diagram along with the experimental data [20,31].

in the Ag-As phase diagram will become wider. Owing to this, the activity data at high contents of As are considered to be inaccurate and given a low weight in the optimization. Thus, the present calculations cannot reproduce well the activity data at high contents of As.

The calculated Ag-Lu phase diagram with the experimental data from Gschneidner et al. [32], and Stepf and Jehn [33] is shown in Fig. 6. The calculated melting temperature of Ag_2Lu is 1098 °C, which is consistent with experimental data [33]. In the present work, the calculated invariant reaction of $L \rightarrow (Ag) + Ag_4Lu$ at 812.7 °C and 11.6 at % Lu is consistent with the reported one [32]. The calculated invariant reactions $L \rightarrow Ag_4Lu + Ag_2Lu$, $L \rightarrow Ag_2Lu + AgLu$ and $L \rightarrow AgLu$

Table 6
Calculated invariant reactions compared with the literature data in the Ag-As system.

Reactions	Type	*T*, (°C)	Composition (at% As)			Reference
L → ζ + (As)	eutectic	540.0	25.3	11.0	100	[31] (exp.)
		531.0	26.0	10.8	100	This work (cal.)
ζ → (Ag) + (As)	eutectoid	446.0	9.5	7.0	100	[31] (exp.)
		448.0	10.0	8.3	100	This work (cal.)
L + (Ag) → ζ	peritectic	582.0	21.5	7.8	10	[31] (exp.)
		589.0	22.6	8.8	9.4	This work (cal.)

exp. = experiment; cal. = calculation.

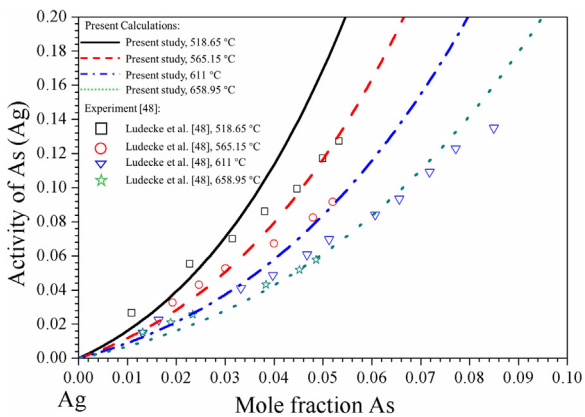

Fig. 5. Calculated activities of As in (Ag) phase compared with the experimental data: 518.65, 565.15, 611 and 658.95 °C [48].

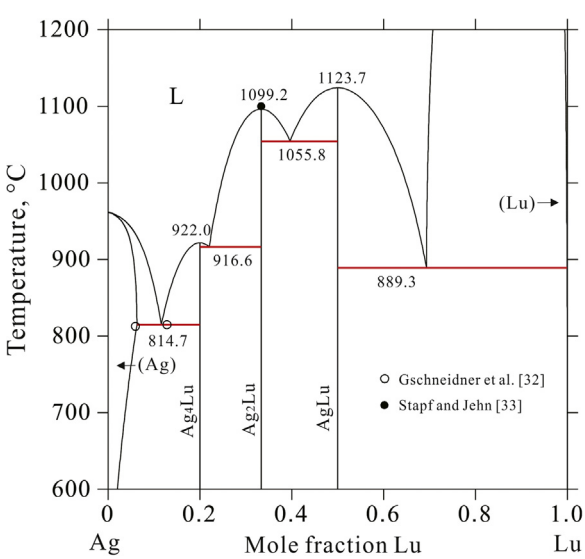

Fig. 6. Calculated Ag-Lu phase diagram along with the experimental data [32,33].

+ (Lu) are at 916.6 °C and 22.0 at% Lu, 1055.8 °C and 37.0 at% Lu, and 889.3 °C and 69.5 at% Lu, respectively. These calculations need further experimental verification. And Table 7 summarizes the computed invariant reactions in the Ag-Lu binary system along with the experimental data [32]. Moreover, the calculated enthalpies of formation of the Ag-Lu system are compared with the literature data, as presented in Fig. 7. As shown in Table 3 and Fig. 7, the calculated enthalpies of formation in the present work are in good agreement with the first-principles calculated values [40,41] and experimental values [52]

Table 7
Calculated invariant reactions compared with the literature data in the Ag-Lu system.

Reactions	Type	*T*, (°C)	Composition (at% Lu)			Reference
L → (Ag) + Ag_4Lu	eutectic	814.0	12.0	5.8	0.2	[32] (exp.)
		814.7	11.6	6.0	0.2	This work (cal.)
L → Ag_4Lu + Ag_2Lu	eutectic	–	–	–	–	–
		916.6	22.0	0.2	0.33	This work (cal.)
L → Ag_2Lu + AgLu	eutectic	–	–	–	–	–
		1055.8	37.0	0.33	0.5	This work (cal.)
L → AgLu + (Lu)	eutectic	–	–	–	–	–
		889.3	69.5	0.5	0.99	This work (cal.)

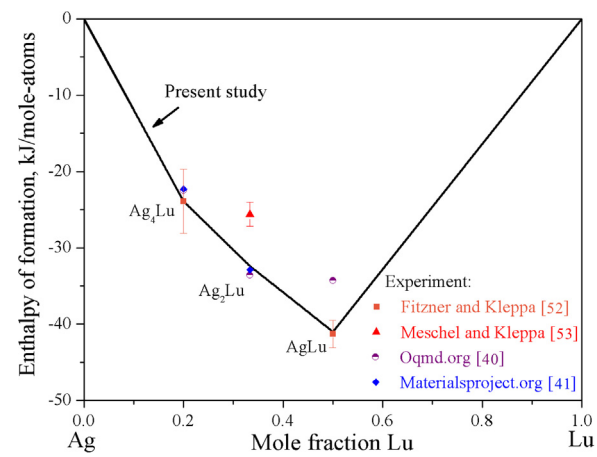

Fig. 7. Calculated enthalpies of formation at room temperature in the Ag–Lu system, compared with the experimental data [40,41,52,53].

except for the values for Ag_2Lu [53] and AgLu [40]. For the Ag_2Lu phase, the presently calculated enthalpy of formation is − 32.4 kJ/mole-atoms, which is lower than the reported one − 25.6 ± 1.6 kJ/mole-atoms [53]. The experimental phase diagram data [33] cannot be reproduced if the experimental enthalpy of formation for the Ag_2Lu phase [53] is used in the optimization. Thus, the experimental data reported by Meschel and Kleppa [53] are given a low weight in the present work. For the AgLu phase, the presently calculated enthalpy of formation is in good agreement with the experimental value [52] and lower than the first-principles computed value [40]. In view of the similarity of the Ag-RE systems [49–51], the AgRE phase has a higher melting point and lower value of enthalpy of formation than the Ag_2RE phase. The AgLu and Ag_2Lu phases have similar enthalpies of formation according to the first-principles calculations value [40]. Thus, the experimental enthalpy of formation [52] of the AgLu phase is more reasonable than the first-principles calculations value [40]. Further experimental measurement on enthalpies of formation of the Ag_2Lu and AgLu is needed to verify the present calculations.

The thermodynamic parameters for the Ag-X (X = *S*, As, Lu) binary systems obtained in the present work can enrich the thermodynamic database for Ag-based alloys, which can provide the phase equilibria and thermodynamic data for designing new Ag-based alloys.

Conclusions

The literature phase diagram and thermodynamic data in the Ag-X (X = *S*, As, Lu) systems were critically reviewed. Based on the critically reviewed literature data, self-consistent thermodynamic parameters were obtained for each binary system by using the CALPHAD approach

for the first time. The calculated phase diagrams and thermodynamic properties are in reasonable agreement with the experimental data. An associate model is used to describe the liquid phase in the Ag-S system. In the Ag-As and Ag-Lu systems, the liquid phases were described by the substitutional solution model.

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